Question	Answer	Marks
9(a)(i)	positron	**B1**
9(a)(ii)	$\lambda = \ln 2 / (110 \times 60) = 1.05 \times 10^{-4}\ s^{-1}$	**A1**
9(a)(iii)	$N = M / (18\ u)$ **or** (M in grams $\times N_A / 18$) $N = (2.1 \times 10^{-12}) / (18 \times 1.66 \times 10^{-27})$ **or** $(2.1 \times 10^{-9} \times 6.02 \times 10^{23}) / 18$ $(= 7.0 \times 10^{13})$	**C1**
	$A = \lambda N$	**C1**
	$= 1.05 \times 10^{-4} \times 7.0 \times 10^{13}$ $= 7.4 \times 10^{9}$ Bq	**A1**
9(b)(i)	• (pair) annihilation occurs • the <u>mass</u> of the two particles is converted into energy • two <u>gamma</u> photons are formed and travel in opposite directions **or** two <u>gamma</u> photons are formed and leave the body • difference in arrival times of photons (at detector) is processed *Any three points, 1 mark each*	**B3**
9(b)(ii)	with a shorter half-life: sample would (almost) fully decay before the test is complete	**B1**
	a longer half-life: exposes patient to harmful/ionising radiation unnecessarily **or** with a longer half-life: a larger dose (of tracer) needed to produce detectable activity	**B1**

Question	Answer	Marks